

Fixing GitHub SSH: Permission denied (publickey)

Cloning via SSH on a Linux server (root user)

Cause: the server user does not have an SSH private key, or GitHub does not recognize the matching public key. Use Option A to fix SSH auth, or Option B for a quick HTTPS clone.

Option A: Configure SSH key authentication

1) Check existing keys

```
ls -la /root/.ssh
```

2) Generate a key (if needed)

```
mkdir -p /root/.ssh  
chmod 700 /root/.ssh
```

```
ssh-keygen -t ed25519 -C "ubuntu-8gb-hel1-1" -f /root/.ssh/id_ed25519  
chmod 600 /root/.ssh/id_ed25519  
chmod 644 /root/.ssh/id_ed25519.pub
```

3) Start ssh-agent and add key

```
eval "$(ssh-agent -s)"  
ssh-add /root/.ssh/id_ed25519
```

4) Trust GitHub host key

```
ssh-keyscan -t rsa,ecdsa,ed25519 github.com >> /root/.ssh/known_hosts  
chmod 644 /root/.ssh/known_hosts
```

5) Copy public key and add to GitHub

```
cat /root/.ssh/id_ed25519.pub
```

6) Test GitHub SSH login

```
ssh -T git@github.com
```

7) Clone the repo/branch

```
git clone -b release_26_01 git@github.com:Lovepreet1984/cms.git
```

Where to add the public key

Best for servers: Repo Settings -> Deploy keys. Alternative: GitHub account Settings -> SSH and GPG keys.

Option B: HTTPS + Personal Access Token (PAT)

Use HTTPS if you want a quick clone. When prompted for a password, use a GitHub PAT.

```
git clone -b release_26_01 https://github.com/Lovepreet1984/cms.git
```

Troubleshooting

If it still fails, run verbose SSH to see which key is offered and why auth fails:

```
ssh -vT git@github.com
```

Security note: Never share your private key (e.g., /root/.ssh/id_ed25519). Sharing the public key (.pub) is safe.